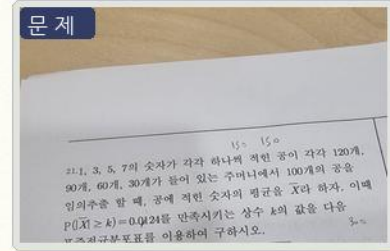


확률과 통계

21번

표준정규분포
표본평균 추론

문제



문제 파악

공 1, 3, 5, 7이 각각 120, 90, 60, 30개 (총 300개)

100개 임의추출 후 표본평균 $\bar{X}$

$$P(|\bar{X} - \mu| \geq k) = 0.0124 \rightarrow k = ?$$

풀이 과정

1. 모평균(μ) 계산

$$P(X=1)=2/5, P(X=3)=3/10, P(X=5)=1/5, P(X=7)=1/10$$

$$\begin{aligned} \mu &= 1 \times (2/5) + 3 \times (3/10) + 5 \times (1/5) + 7 \times (1/10) \\ &= 4/10 + 9/10 + 10/10 + 7/10 = 30/10 = 3 \end{aligned}$$

2. 모분산(σ^2) 계산

$$E(X^2) = 1 \times (2/5) + 9 \times (3/10) + 25 \times (1/5) + 49 \times (1/10) = 13$$

$$\sigma^2 = E(X^2) - \mu^2 = 13 - 9 = 4 \quad (\sigma = 2)$$

3. 표본평균의 분포 ($n=100$)

$$\sigma(\bar{X}) = \sigma / \sqrt{n} = 2/10 = 0.2$$

$$\bar{X} \sim N(3, 0.2^2) \quad \langle \text{중심극한정리} \rangle$$

4. k 계산

$$P(|\bar{X} - 3| \geq k) = 0.0124$$

$$P(|Z| \geq k/0.2) = 0.0124$$

$$2 \times P(Z \geq k/0.2) = 0.0124 \Rightarrow P(Z \geq k/0.2) = 0.0062$$

$$P(0 \leq Z \leq k/0.2) = 0.5 - 0.0062 = 0.4938$$

$$\text{표에서 } Z=2.5 \Rightarrow P(0 \leq Z \leq 2.5) = 0.4938 \quad (V)$$

$$\text{따라서 } k/0.2 = 2.5 \Rightarrow k = 0.5$$

$$k = 0.5 = 1/2$$

핵심 공식

표본평균: $\bar{X} \sim N(\mu, \sigma^2/n)$

표준화: $Z = (\bar{X} - \mu) / (\sigma / \sqrt{n})$

양측확률: $P(|Z| \geq z) = 2 \times (0.5 - P(0 \leq Z \leq z))$